Shayan Salehi

+98 913-809-4457 s.salehi1381@gmail.com ShayanSalehi81 LiknedIn Google Scholar

Education

M.Sc. | *Artificial Intelligence* Sep 2025 – Present
Sharif University of Technology Tehran, Iran
- Overall: 18.56/20.00

B.Sc. | *Computer Engineering* Sep 2020 – Feb 2025
Sharif University of Technology Tehran, Iran
- Overall: 18.90/20.00 • Major: 18.96/20.00

Research Interests

- Vision Language Models
- Tool-Augmented Language Models
- Biomedical Informatics
- Human-Computer Interaction
- AI in Healthcare

Publications

Learning from Disagreement: Maximum Divergence Knowledge Distillation Published
A. Jafari, P. Fashi, M. Rezagholizadeh, H. Golmankhaneh, **S. Salehi**, V. Appia, E. Barsoum, A. Ghodsi
Published in the NeurIPS 2026 Conference.

The Compute Floor of Data Mixing: When Loss-Based Reweighters Start to Drain Easy Synthetic Text Published
S. Salehi, I. Mohammadi
Published in the COLM 2026 Workshop on Methods and Opportunities at Small Scale (MOSS).

PrivacyPeek: Showing Users What Their AI Chats Send Out, and When It Crosses a Line Published
I. Mohammadi, **S. Salehi**
Published in the USIT 2026 Poster Session.

Structure Matters: Evaluating Multi-Agents Orchestration in Generative Therapeutic Chatbots Published
S. Elahimanesh, M. Mohammadkhani, S. Zahedi Movahed, M. Abootorabi, **S. Salehi**, A. Edalat
Published in the ACM CUI 2026 conference.

Folk Theories of Federated Moderation: How Users Make Sense of Cross-Instance Action Published
I. Mohammadi, **S. Salehi**
Published in the CSCW 2026 Poster Session.

Understanding Students' Rationales for Cheating with AI: An Instructor Perspective Published
I. Mohammadi, S. Rismanchian, **S. Salehi**
Published in the ICLS 2026 Conference Poster Session.

From Words and Exercises to Wellness: Farsi Chatbot for Self-Attachment Under Review
S. Elahimanesh, **S. Salehi**, S. Zahedi Movahed, L. Alazraki, R. Hu, A. Edalat
Submitted to the EAI Endorsed Transactions on Pervasive Health and Technology journal.

ToolRacer: Retrieval-Aware Co-Training and Error Recovery for Small Language Model Tool Calling In Progress
S. Salehi, M. Bagherifard, E. Mohajeri

Can Medical VLMs Reason from Degraded Evidence? A Paired Microscopy Robustness Benchmark In Progress
S. Salehi, I. Mohammadi

Vision-Language Models in Radiology: Improving Trust with Fact-Based Report Generation In Progress
S. Salehi, A. Salesi, S. Daneshgar, S. Beyrami

Honors and Awards

National M.Sc. Entrance Exam Feb. 2025
Ranked 16th among 22,000+ candidates (Top 0.07%)

National B.Sc. Entrance Exam Oct. 2020
Ranked 25th among 155,000+ candidates (Top 0.02%)

National Student Olympiad Aug. 2024
Ranked 10th among 1,200+ candidates (Top 0.8%)

NODET Entrance Exam Sep. 2014
Ranked top 120 among 11,000+ candidates (Top 1.1%)

NLP Tech Olympics 2025 Oct. 2025
Ranked 1st among 2100+ teams (Top 0.05%)

Research Experience

Maximum Divergence Knowledge Distillation | ***University of Waterloo*** July 2025 – May 2026
Supervised by Prof. Ghodsi - Remote Collaboration

- Proposed **MDKD**, a rejection-sampling distillation method targeting regions where student LLMs underestimate the teacher.
- Achieved state-of-the-art performance on **GSM8K**, summarization, and code generation.

Mental Wellness Chatbot | ***Imperial College London*** April 2023 – February 2026
Supervised by Prof. Edalat - Remote Collaboration

- Developed a **chatbot** recommending exercise regimens using **self-attachment theory (SAT)** protocols.
- Trained transformer-based **emotional classifier** in a range of 12 classes with a 92% f1 score.
- Used **PPO reinforcement learning** techniques to fine-tune the GPT-2 model for enhanced dialogue generation.
- Worked on **RAG-based LLM systems** alongside fine-tuning the **Llama-3** model using mental health datasets.

Synthetic Data Analysis on Data Selection Methods | ***Rensselaer Polytechnic Institute*** April 2025 – February 2026
Supervised by Prof. Amiri - Remote Collaboration

- Implemented an **adversarial approach** to data selection for pretraining foundation models.
- Implemented analyzing methodology of synthetic-data weighting distributions in mixture methodologies such as **DoReMI**.

News Importance Classifier with LLMs | ***Sharif University of Technology*** May 2024 – December 2024
Supervised by Prof. Jafari

- Used **symbol tuning** methodology to evaluate LLM sensitivity to prompt variations.
- Compared **reasoning abilities** of available open-source LLMs based on a novel human-labeled Persian news dataset.
- Designed a **query-dependent prompt system**, maximizing the effectiveness of LLMs across diverse scenarios.

LLMs Citation Benchmark | ***Sharif University of Technology*** June 2024 – October 2024
Supervised by Prof. Asgari

- Developed and implemented an exhaustive framework for evaluating the **medical citation quality** of LLMs.
- Used a **fact-validation** method and weighted scoring system for accuracy and relevance.
- Gathered medical context-related datasets containing texts alongside their **quantized facts**.

Survey on Diffusion Models | ***Sharif University of Technology*** November 2023 – September 2024
Supervised by Prof. Rohban and Prof. Soleymani

- Examined **compositional challenges** during the generation phase of **diffusion models**.

Coursework

Deep Learning (MSc.)	19.7/20.0	Generative Models (MSc.)	19.2/20.0
Natural Language Processing (MSc.)	19.6/20.0	Intelligent Analysis of Bio Images (MSc.)	19.1/20.0
Fundamentals of 3D Computer Vision	20.0/20.0	Machine Learning	20.0/20.0
Linear Algebra	20.0/20.0	French General	20.0/20.0

Head Teaching Assistant

Compiler Design - Dr. S. Hosseinmardi Fall 2023 and Spring 2024
Mobile Programming - Dr. S. Nari Spring 2023

Teaching Assistant

Deep Learning (MSC.) - Dr. M. Soleymani	Fall 2026
Advanced Vision Computer (MSC.) - Dr. S. Kasaei	Fall 2026
Natural Language Processing (MSC.) - Dr. M. Samiei	Fall 2026
Modern Information Retrieval - Dr. M. Soleymani	Spring 2026
Machine Learning - Dr. F. Salehi	Spring 2024
Mathematics for Artificial Intelligence - Dr. A. Najafi	Spring 2024
Game Theory - Dr. M. Seddighin	Spring 2024
Design of Algorithms - Dr. M. Abam	Spring 2024
Probability and Statistics - Dr. M. Jafari	Spring 2024
Machine Learning - Dr. A. Motahari	Spring 2024 and Fall 2023
Database Design - Dr. M. Akhi	Fall 2023
Discrete Structures - Dr. M. Abam	Summer 2023
Data Structures and Algorithms - Dr. M. Ghorbani	Spring 2023
Numerical Methods - Dr. S. Hosseinghorban	Fall 2023, Spring 2023, and Fall 2022
Probability and Statistics - Dr. M. Motahari	Fall 2022
Linear Algebra - Dr. S. Hosseinghorban	Spring 2023, Fall 2022, and Spring 2022
Advance Programming - Dr. V. Salmani	Fall 2021

Work Experience

Senior ML Engineer June 2025 – Present

AVIA

- Served and benchmarked **system one** models like JEV on decision-making tasks.
- Implemented **LiveKit**-based backend for **VoIP** and AI voice assistant services.
- Developed end-to-end chat-session evlautor dashboard using **Django**, **Typescript**, and **React**.
- Enhanced **prompt and prefix caching** system for served LLM models.

LLM Engineer October 2025 – June 2026

MCINEXT

- Implemented **router model** for intent and tool selection for **dynamic prompt** generation.
- Designed and implemented **AR-GRPO** training pipeline for fine-tuning agentic LLMs such as **Gemma3-4b** and **Qwen3-4b**.
- Developed **Search Engine evaluation** system and wrote proper CICD and Helm Charts to serve the services on **ArgoCD**.
- Created **adaptive agentic** LLM/Tool-Retriever system flow architecture.
- Implemented end-to-end chatbot evaluation system using **LLM-based evaluation**, win-rate metrics, and **Django** framework.

AI Engineer February 2025 – September 2025

Mohaymen ICT Group

- Maintained AI-related production services via **Docker Compose** and implemented 3-phase chat response caching using **Redis**.
- Boosted RAG-system performance through **asynchronous programming** and **multi-process logic**.
- Implemented end-to-end **NER detection** with ParsBERT, **translation** with SeamlessM4T all deployed via **FastAPI**.
- Designed **ASR hybrid search** service using NeMo model and **OCR pipelines** using Tesseract, RolmOCR models, and FastAPI.
- Integrated log and chat tracing with **Phoenix** and utilized **vLLM** for offline LLM inference.
- Conducted load testing of upstream and production systems using **Locust**.

Machine Learning Engineer July 2024 – November 2024

Mohaymen ICT Group

- Fine-tuned a **BERT model** for improved detection of similar questions to the user's query.
- Developed an **intelligent spelling mistake corrector** based on context, frequency analysis, and similarity score.
- implemented **RAG-based LLM system** by using **LangChain** and **LangGraph** libraries.
- Enhanced **speech-to-text** and **text-to-speech** model accuracy using the **NeMo** framework.

Software Engineering Intern June 2021 – September 2021

Code Star Internship

- Designed and developed efficient **data pipelines** utilizing C# to ensure streamlined data flow and processing.
- Gained insight of **database fundamentals**, effectively leveraging SQL for data extraction, transformation, and loading operations.

Projects

AYA Prompt-Based Classification | ***Bachelor Project*** | Github May 2024 - Present

- Engineered an automated system for classification tasks utilizing K-shot learning and TF-IDF similarity.
- Integrated fine-tuned Aya23 model with quantized and LoRA techniques.

RAG Based ChatBot | ***Star Division*** | Github July 2024 – September 2024

- Designed a chatbot framework with context-based question answering, spelling correction, and profanity filtering.
- Leveraged LangChain and OpenAI API library, incorporating retrieval-augmented generation for enhancing the accuracy.
- Improved response relevance by using FAISS vector store with text embeddings for enhanced performance.

Medical Question Answering System | ***Natural Language Processing*** | Github May 2024 – June 2024

- Fine-tuned T5-based models on the PubMed dataset to generate precise medical answers.

Film Genre Detection | ***Natural Language Processing*** | Github April 2024 – May 2024

- Devised models for film genre classification using both traditional machine learning and transformer-based models.
- Conducted named entity recognition (NER) on Persian and English synopses, utilizing the CONLL dataset.

Feature Matching | ***3D Computer Vision*** | Github January 2024 – February 2024

- Developed a tool for feature detection and matching across images, incorporating classic and deep learning approaches.

SAT Teacher | ***Imperial College London Collaboration*** | Github June 2023 – September 2023

- Created a QA tool using BERT embeddings to enhance user understanding of self-attachment techniques (SAT).

Image Captioning | ***Machine Learning*** | Github February 2023

- Constructed an LSTM encoder-decoder architecture for automatically generating descriptive captions for images.

CMinus Compiler | ***Compiler Design*** | Github September 2022 – January 2023

- Programmed a compiler for CMinus Language, a simplified version of C, using Python and its standard libraries.
- Developed the Lexical Analyzer, Parser, Code Generator, and Semantic Analyzer, completing a full compilation pipeline.

Community Involvement

Karsoogh Mehregan 23 & 24 & 25 & 26 | ***NODET Alumni Community*** August 2024, 2024, 2025

Head of Education & Tournament

- Served as the scientific and tournament team leader for the event.
- Hosted a cryptography and artificial intelligence workshop for middle school students.
- Designed various online games to teach ML-learning, reinforcement learning, and cryptography concepts.

WebMasters | ***Sharif University of Technology*** April 2022

Mentor

- Hosted a free community tutorial on front-end programming for local students.
- Provided a wide range of HTML, CSS, and JavaScript lessons.

Skills

DevOps: Docker, Docker Compose, Kubernetes, Redis, ArgoCD, Kibana
Programming: Python, Java, C#, PostgreSQL, MongoDB, Swift, FastAPI, RestAPI, Django
Web Programming: HTML, CSS, JavaScript, TypeScript, React
Document Creation: LaTex, Markdown
Testing/Monitoring: Locust, Phoenix
Data Science Libraries: Pytorch, Tensorflow, Sklearn, Transformers, Matplotlib, Numpy, Pandas, Streamlit, Rasa, Faiss, vLLM **Research and Reference Management**: Zotero, Citavi

Languages

English: Professional proficiency

- ***IELTS*** Overall: 7.5
  Reading: 8.5, Listening: 8.0, Speaking: 7.0, Writing: 7.0

French: Elementary proficiency
Arabic: Intermediate proficiency
Persian: Native proficiency